

Appendix A2: conditional probability tables.

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Network structure

Explanations

Decision nodes

Controls the climate and fishing management scenarios, those are defined a priori depending on the conditions to be explored in the scenario analyses.

Extreme weather events (node: event)

The node *Event* is used to wrap all the extreme weather events we are considering.

Levels (and acronyms)

- **Storms** (sto). Strong wind events, with intensity above 24.5 m/s. Season: summer.
- **Gales** (gal), or moderately strong wind events, with intensity between 15 m/s and 24.5 m/s;
- **Extreme precipitation** (erf) – rainstorm (summer), or deviation from average precipitation regime;
- **Algal bloom** (abl): intense algal bloom;
- **Summer heatwaves** (hws): prolonged discrete anomalously warm water event (Hobday et al., 2016);
- **Winter heatwaves** (hww): prolonged discrete anomalously warm water event (Hobday et al., 2016);
- **Icing** (Ici): strong wind events associated to ice, which then forms icing pile up on gears and boats.

Fishing style

The node *Fishing style* is used to wrap all the fishing styles and seasons.

Levels (and acronyms)

- **Recreational fisher in winter season** (rew): a fishing style that primarily performs ice fishing and potentially can fish in open water from the shore;
- **Recreational fisher in summer season** (res): a fishing style that primarily performs ice fishing and potentially can fish in open water from the shore;
- **fishing guide in winter season** (fgw), a fishing style that primarily performs ice fishing and potentially can fish in open water from the shore;
- **fishing guide in the summer season** (fgw), which primarily fish in coastal areas from shore or boat and can potentially move to inland fishing;
- **Coastal small-scale fishers**, (ssf) a fishing style that happen mostly in summer using gillnets in coastal areas, and in some cases with traps (not for salmon) but can perform limited amount of ice fishing in winter;
- **Coastal small-scale fishers specialised in salmon fishing** (sal): which fish for salmon with fixed traps rigidly regulated;
- **Pelagic trawlers** (trw), represented by pelagic trawlers mostly active in winter and exclusively fishing for small pelagic;
- **Archipelago fishers** (arc), which is similar to an artisanal fishers but is by law prevented from selling their catches.

Fishing management

it controls how much fishing regulations allows fishers to implement adaptation strategies (e.g.: changing target species and gear).

Levels (and acronyms)

- **Business as Usual** (BAU): each fishing style adopts strategies and behaviour according to interviews results
- **Flexible management** (FLEX). Fishing style is allowed for more flexibility.

Stock status

it controls how much fishing regulations allows fishers to implement adaptation strategies (e.g.: changing target species and gear).

Levels (and acronyms)

- **status quo** (SQ): each fishing style adopts strategies and behaviour according to interviews results
- **improved stock status** (FLEX). Fishing style is allowed for more flexibility.
- **worsened stock status**

Operational subsystem

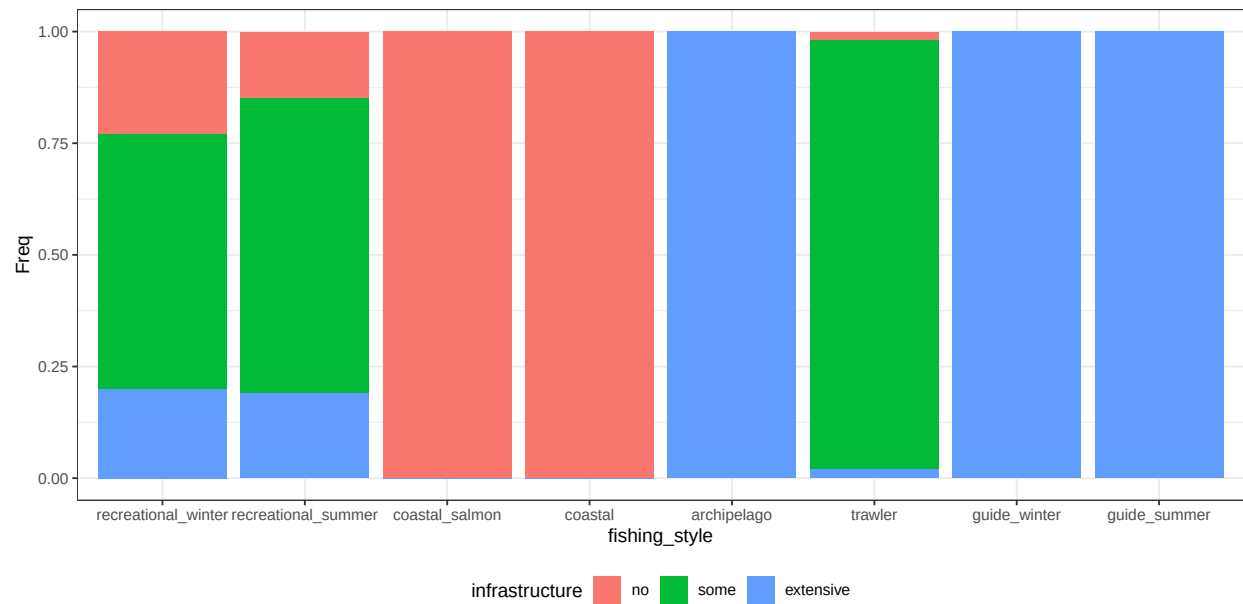
Infrastructure

Parents: none

Method: M1 based on Q17

Definition: availability of terrestrial infrastructures such as roads, ports and bridges.

Levels: no; some; extensive



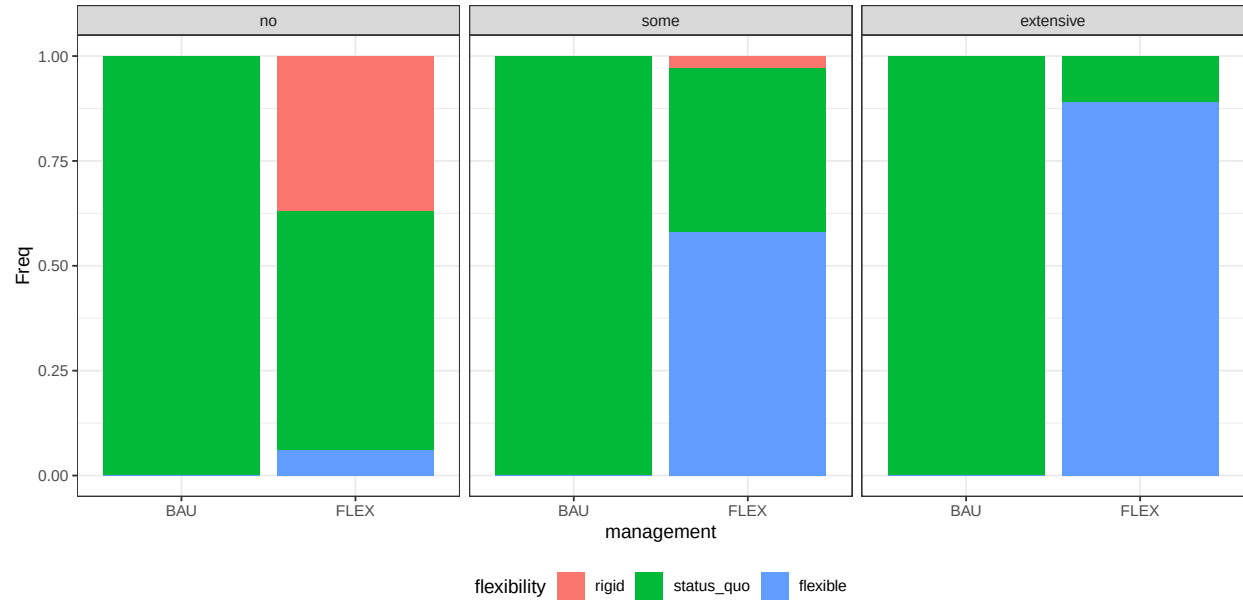
Flexibility

Parents: infrastructure, management

Method: M3

Definition: choosing different flexibility possibility to change fishing strategy (both gear and target).

Levels: rigid; status quo; flexible



Strategy

Parents: flexibility, event, fishing style

Method: M3. Each interviewee expressed their view regarding the strategy in different questions. For each event and fisherman, it is applied manual revision to assign one or more between react, cope and adapt. Then it is identified which answer is more appropriate to fill the CPT. Case 1: single strategy. React use avoidance (or what has been labelled as such) to define the probability to cope or react. Adapt use avoidance (or what has been labelled as such) to define the probability to be cope or adapt. Cope or NA: then is 100% cope. When there is a single respondent use the self-declared uncertainty. When there are multiple respondents (which all agree on the strategy), calculate uncertainty from answers. Case 2: single strategy but multiple options (i.e.: target species). Choose the highest value, the range of values would be used to describe the node target conditional on event and strategy. Case 3: When multiple strategies occur within the same context, the probabilities are combined through a weighted noisy-OR mechanism, accounting for both response frequency and uncertainty. This means that first we separately calculate the probability distribution as if it was case 1, then the probabilities are combined as weighted average where the weight is the number of responses.

Definition: Fishers response is coded as Adapt (proactive planning); react (unplanned response); cope (accept passively) (Green et al., 2021).

Levels: Cope is when fisher prefer to not go out fishing; React is when they apply small modification to their practice, such as fishing in deeper areas or in different time of the day; Adapt is when they take proactive response such as modify their strategy by changing fishing target and/or gear.

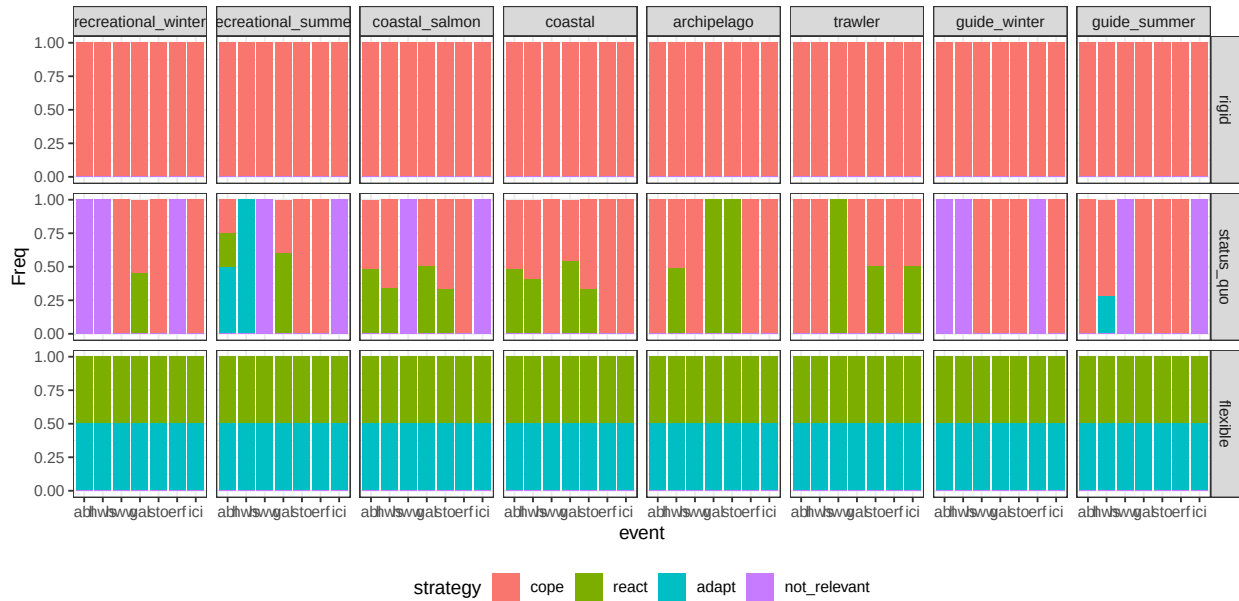


Figure A2.1: strategy by fisherman

Technical solutions

Parents: strategy, event, fishing style

Method: The CPT is compiled manually for each combination of event and fisherman based on what has been mentioned throughout the questionnaire. In case only one strategy is mentioned, this is assigned probability = 1. When two strategies are mentioned, they are 50% probable unless one strategy is explicitly adopted more often than the other (and so on for three strategies).

Definition: measures implemented in the case when the strategy is React. The different solutions work differently and have different cost.

Levels: travel further implies fisherman change fishing location to seek a sheltered place or to find deeper waters; short sets mean that the fisher perform shorter sessions than average (i.e.: reduced soak time for gillnets); other technical includes strategies that are mentioned sporadically (i.e.: use of ice machines).

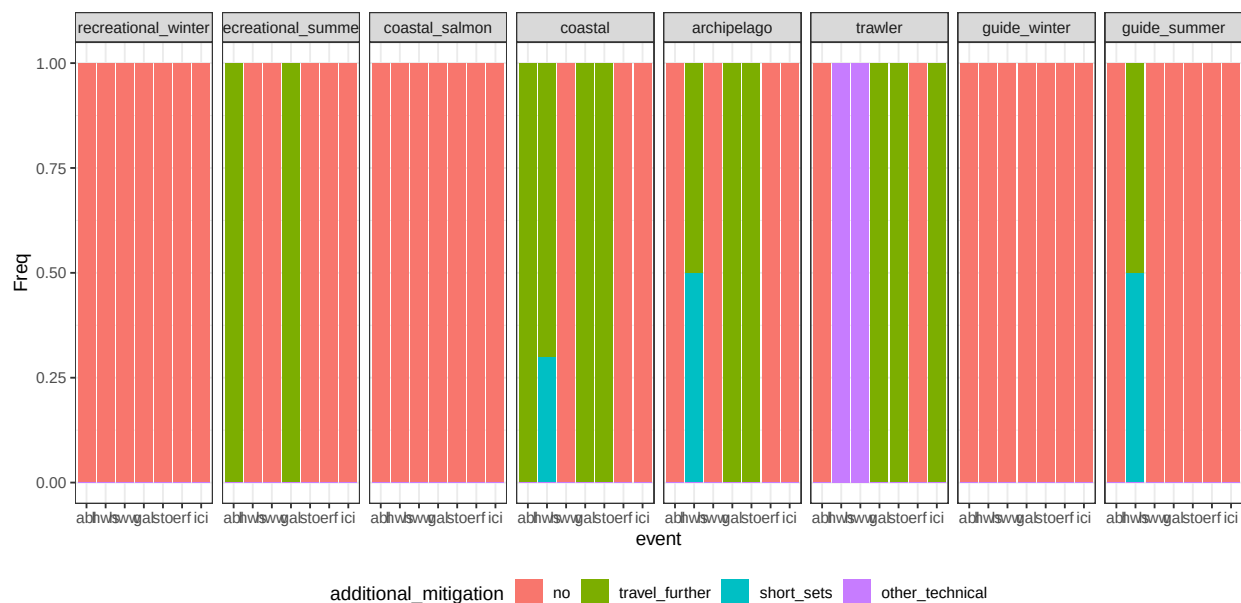


Figure A2.2: additional_mitigation by fisherman

Gear

Parents: strategy, event, fishing style

Method:

Definition:

Levels:

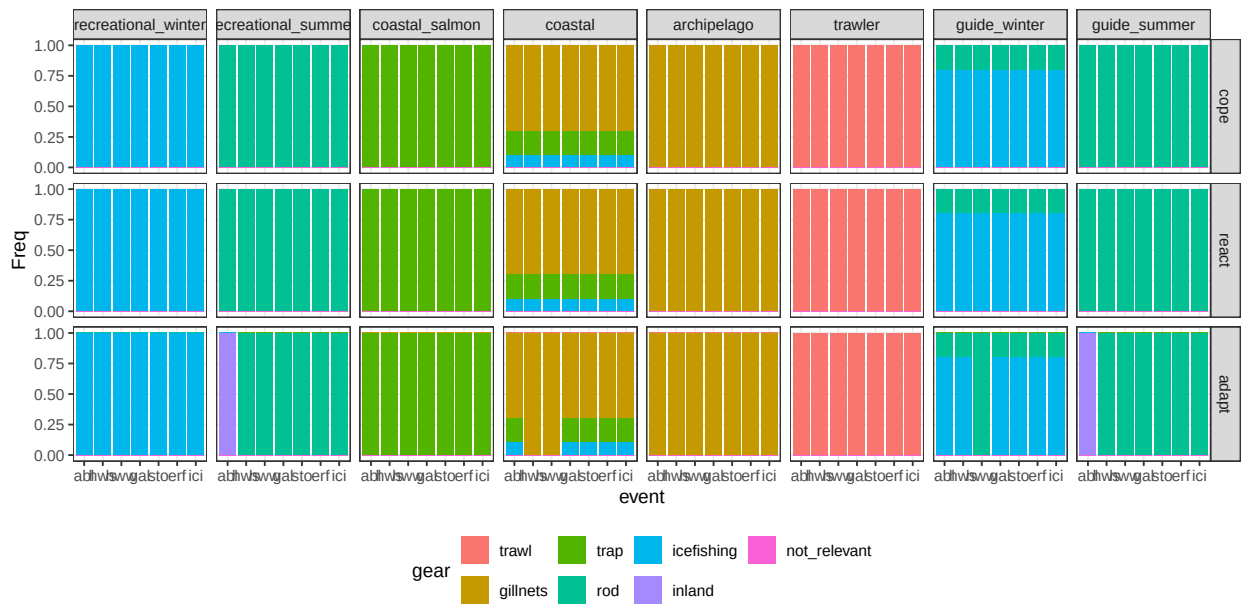


Figure A2.3: gear by fisherman

Target

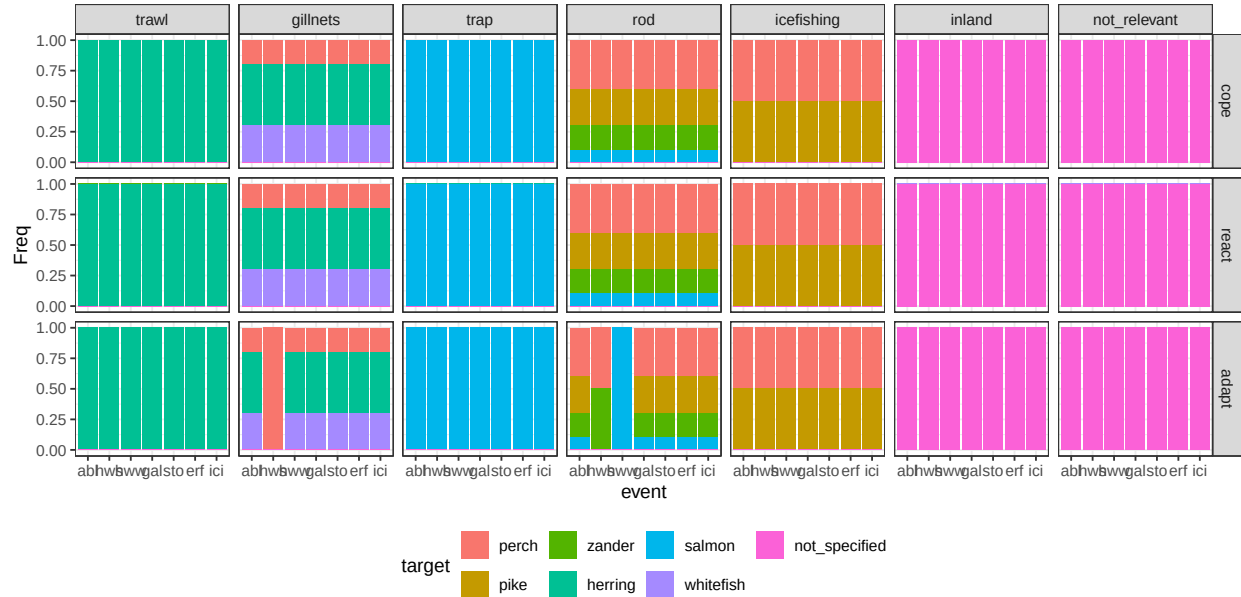
Parents: gear, strategy, event. *Levels:* perch, pike, zander, herring, salmon, whitefish, not_specified.

Parents: gear, strategy, event

Levels: perch, pike, zander, herring, salmon, whitefish, not_specified

Method:

Definition:



Personal safety

Parents: technical solutions, event, gear

Method:

Definition:

Levels:

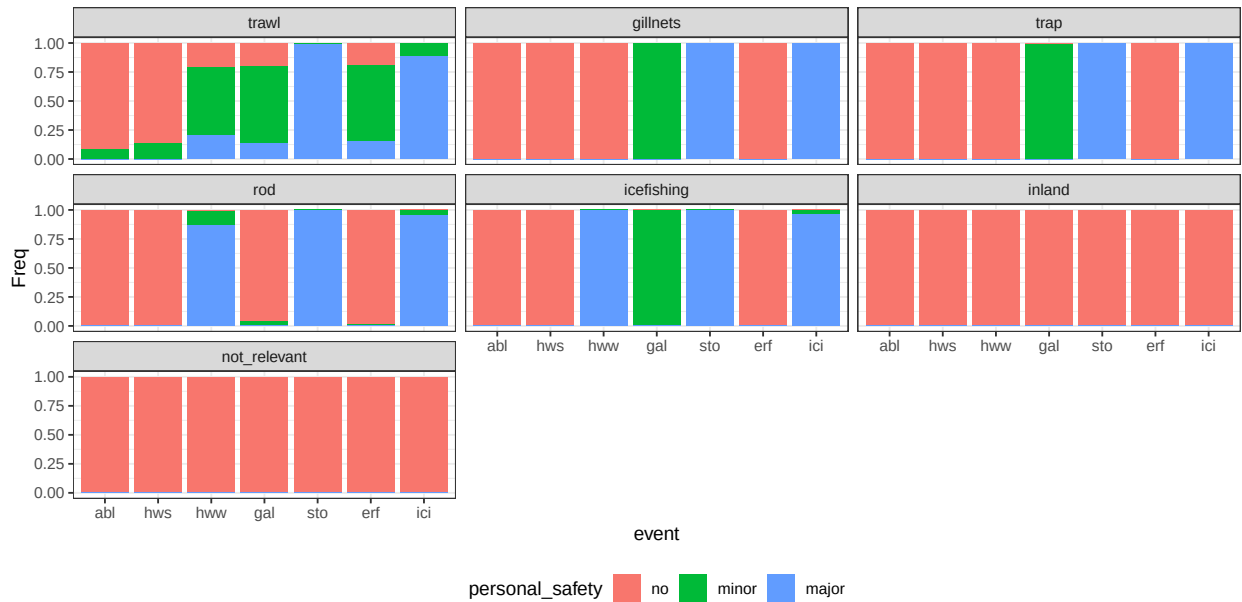


Figure A2.4: gear by fisherman